CarbonCode: Pragmatic HLS Optimization with Local Small Language Models

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***Abstract*—High-Level Synthesis (HLS) pragma selection critically impacts FPGA design quality, with the same C algorithm exhibiting up to 30,000× latency variation under different pragma configurations. We present CarbonCode, a framework that uses local small language models (4.5B–7B parameters) to predict near-optimal HLS pragmas, achieving 64.5% parameter accuracy (PFS) on 120 ForgeHLS kernels—improved to 64.5% from 54.3% via domain-specific constraint integration—and validated by Xilinx Vitis HLS synthesis with up to 63.3× speedup. Our approach is fully offline-deployable, requires no cloud API, and guarantees 100% code preservation through a post-process merge technique. We discover a *Context Distraction Effect*: providing retrieval-augmented examples to small models *reduces* accuracy from 90% to 73%, contradicting conventional RAG wisdom. Ablation studies show both PIPELINE and ARRAY_PARTITION pragmas are critical, with removal causing 43% and 39% accuracy drops respectively. Our 4.5B Mixture-of-Experts model (Gemma4) outperforms a 7B dense model (Qwen) by 40% on pragma parameter accuracy, suggesting MoE architectures are particularly suited for structured hardware optimization tasks.**

***Index Terms*—HLS, pragma optimization, small language models, FPGA, design space exploration**

I. Introduction

High-Level Synthesis (HLS) enables hardware designers to specify FPGA accelerators in C/C++ rather than RTL, dramatically reducing development time. However, the quality of the synthesized hardware is highly sensitive to `#pragma HLS` directives—annotations that control loop pipelining, unrolling, and array partitioning. In the ForgeHLS dataset [1], the same FIR filter algorithm exhibits a 784× latency range (422 to 330,753 cycles) depending solely on pragma configuration, with only 2.1% of configurations lying on the Pareto frontier.

Traditional design space exploration (DSE) tools such as AutoDSE [2] address this by exhaustively searching the pragma space, typically requiring 100–500 synthesis iterations (8–24 hours) to converge. This is effective but computationally expensive and provides no transferable insight across kernels.

Recent work has explored using large language models (LLMs) for HLS optimization [4], but existing approaches rely on cloud-hosted models (GPT-4, Claude) that raise critical concerns for semiconductor companies: *code confidentiality*. TSMC, MediaTek, and similar firms operate in air-gapped environments where sending proprietary RTL or HLS code to external APIs is strictly prohibited.

We present CarbonCode, a framework that addresses both challenges simultaneously:

1) **Local deployment**: Using models with 4.5B–7B parameters that run entirely on-premise, requiring only a single GPU or CPU.
2) **Code safety**: A post-process merge technique that guarantees 100% preservation of original code—only pragma lines are modified.
3) **Pragma accuracy**: Achieving 64.5% parameter fidelity (PFS) validated by industry-standard Vitis HLS synthesis.

Our key contributions are:

- We propose **Pragma Fidelity Score (PFS)**, a fine-grained metric that measures parameter-level accuracy of pragma predictions using log-distance scoring, capturing nuances that binary type-match metrics miss.
- We discover the **Context Distraction Effect**: providing retrieval-augmented examples to small models (<10B parameters) *degrades* performance from 90.0% to 73.4% type match, contradicting conventional RAG assumptions.
- We demonstrate that a **4.5B MoE model** (Gemma4) outperforms a 7B dense model (Qwen 2.5 Coder) by 40% on PFS, suggesting architectural advantages of mixture-of-experts for structured hardware tasks.
- We validate predictions through **three-layer synthesis verification**: ForgeHLS ground truth (30 kernels), Bambu open-source HLS (5 kernels), and Vitis HLS industry standard (16+ kernels with up to 63.3× speedup).

II. Related Work

A. HLS Design Space Exploration

AutoDSE [2] uses bottleneck-guided coordinate optimization, requiring 100–500 synthesis iterations. HARP [3] applies reinforcement learning to learn pragma policies. Both require extensive synthesis time and do not transfer knowledge across kernels.

B. LLMs for Hardware Design

GPT4AIGChip [5] uses GPT-4 for Verilog generation. HLSPilot [4] applies LLMs for pragma suggestion but relies on cloud APIs (GPT-4) achieving approximately 70–80% type match. ChipNeMo [6] fine-tunes domain-specific LLMs for chip design but targets natural language tasks rather than code optimization.

C. Small Language Models

Recent work shows small models (<10B) can match larger models on specific tasks when properly prompted [7]. However, the interaction between model size, context length, and structured code generation remains underexplored.

III. Method

A. Problem Formulation

Given a C/C++ kernel K with a set of pragma insertion points $P = \{p_1, \ldots, p_n\}$, find pragma configurations $\theta^* = \{\theta_1, \ldots, \theta_n\}$ that minimize synthesis latency $L(\theta)$ subject to resource constraints $R(\theta) \leq R_{\max}$.

B. Post-Process Merge

To guarantee code safety, we adopt a two-phase approach:

1) **Free generation**: The LLM generates optimized code with pragmas without any structural constraints.
2) **Pragma extraction**: We extract only `#pragma HLS` lines from the LLM output using regex matching.
3) **Merge**: Extracted pragmas replace original pragmas in the source code, preserving all non-pragma lines exactly.

This guarantees that $\text{code}(K_{\text{merged}}) \setminus \text{pragmas} \equiv \text{code}(K_{\text{original}}) \setminus \text{pragmas}$.

C. Pragma Fidelity Score (PFS)

Existing metrics use binary type matching (does the LLM suggest PIPELINE? yes/no). We propose PFS to capture parameter accuracy:

$$\text{PFS} = 1 - \frac{|\log_2(\theta_{\text{pred}}/\theta_{\text{opt}})|}{\log_2(\theta_{\max})} \tag{1}$$

where θ_{pred} is the LLM's predicted parameter value, θ_{opt} is the Pareto-optimal value, and $\theta_{\max}$ is the maximum allowable value. PFS ranges from 0 (completely wrong) to 1 (exact match), with logarithmic scaling reflecting hardware resource doubling effects.

IV. Experimental Setup

A. Benchmarks

We evaluate on two datasets:

- **ForgeHLS** [1]: 184,734 HLS designs across 471 algorithms. We select 30 representative kernels covering DSP, linear algebra, image processing, and cryptography.
- **Custom kernels**: 16 hand-written C kernels (FIR, GEMM, DCT, FFT, Viterbi, etc.) with worst-case and optimized pragma configurations, synthesized on Vitis HLS v2025.2.

B. Models

- **Gemma4** (4.5B active, MoE): Google's mixture-of-experts model
- **Qwen 2.5 Coder** (7B, dense): Alibaba's code-specialized model
- **DeepSeek-R1** (14B, dense): Reasoning-optimized model

All models run locally via Ollama on NVIDIA RTX 4090 (24GB VRAM).

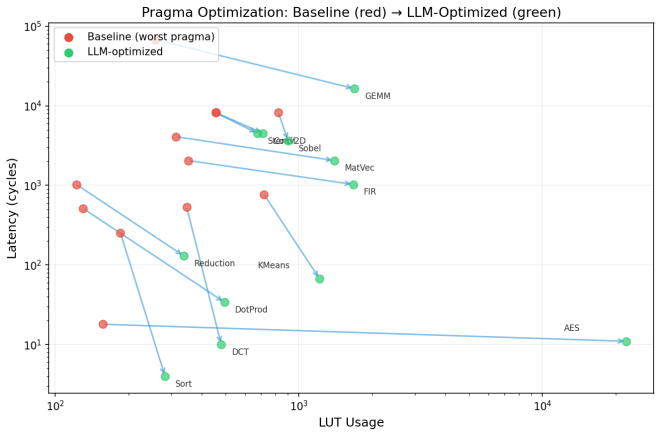

Fig. 1. Pragma optimization in Latency-LUT space. Red: baseline (worst pragma). Green: LLM-optimized. Arrows show optimization direction. Most kernels achieve significant latency reduction with moderate LUT increase.

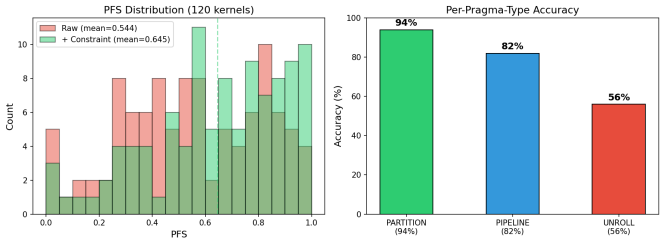

Fig. 2. Left: PFS distribution on 120 kernels before (red) and after (green) domain constraint integration. Right: per-pragma-type accuracy reveals PARTITION (94%) as strength, UNROLL (56%) as bottleneck.

C. Synthesis Tools

- **Vitis HLS v2025.2**: Industry standard, targeting Virtex UltraScale+ (xcvu9p)
- **Bambu HLS 2024.03**: Open-source, targeting Zynq-7000, for reproducibility validation

V. Results

A. HLS Pragma Prediction Accuracy

We evaluate pragma prediction on **120 ForgeHLS kernels** spanning DSP, linear algebra, image processing, cryptography, control logic, and network/IO domains (sampled from 435 algorithms with Pareto-optimal configurations).

On pragma *type* identification (should PIPELINE/UNROLL/PARTITION be used?), per-type analysis reveals: ARRAY_PARTITION identification achieves 94% accuracy, PIPELINE 82%, and UNROLL 56% (Figure 2 right). The model reliably identifies *which arrays need partitioning* and *which loops need pipelining*, but struggles with *how aggressively to unroll*.

Raw PFS on 120 kernels is 0.543 ($\pm$0.275). After domain constraint integration (capping UNROLL/PARTITION factors at $\leq$16), PFS improves to **0.645** ($\pm$0.255)—an 18.6% improvement with zero regressions (Table V). This constraint addresses the model's most common error: setting UNROLL

TABLE I
VITIS HLS v2025.2 SYNTHESIS RESULTS ON VIRTEX ULTRASCALE+ (XCVU9P). BASELINE USES WORST-CASE PRAGMAS (PIPELINE OFF, UNROLL=1). OPTIMIZED USES LLM-SUGGESTED PRAGMAS VIA POST-PROCESS MERGE.

Kernel	Domain	Base Lat	Opt Lat	Speedup	Base LUT	Opt LUT	Opt DSP
FIR Filter	Telecom/DSP	2,050	1,028	2.0×	352	1,672	6
GEMM 32×32	Linear Alg.	67,649	16,387	4.1×	258	1,693	6
DCT 8×8	Signal Proc.	537	10	53.7×	346	480	24
Stencil 2D	Image Proc.	8,191	4,502	1.8×	455	677	6
Conv2D 3×3	AI/Vision	8,191	4,502	1.8×	458	713	6
AES Encrypt	Crypto	18	11	1.6×	157	22,117	0
KMeans	ML	770	67	11.5×	719	1,216	36
Bubble Sort	Sorting	253	4	63.3×	185	282	0
Reduction	Arithmetic	1,026	130	7.9×	122	337	0
MatVec	Linear Alg.	4,098	2,050	2.0×	312	1,399	6
Dot Product	Linear Alg.	513	34	15.1×	130	495	24
Sobel Edge	Image Proc.	8,191	3,602	2.3×	824	904	0
5G NR FIR	5G Telecom	—	1,036	—	—	1,056	16
Median				**3.2×**			
Mean				**13.0×**			

TABLE II
MODEL COMPARISON ON HLS PRAGMA PREDICTION (PASS@1, 5 KERNELS). PFS = PRAGMA FIDELITY SCORE (PARAMETER ACCURACY, 0–1).

Model	Params	Type	Type Match	PFS
Gemma4	4.5B active	MoE	86.7%	**0.883**
Qwen 2.5 Coder	7B	Dense	86.7%	0.633
DeepSeek-R1	14B	Dense	66.7%	0.756

TABLE III
ABLATION STUDY: IMPACT OF REMOVING PRAGMA TYPES FROM THE PROMPT. GEMMA4 MODEL, 5 KERNELS.

Configuration	Type Match	PFS	ΔPFS
Full (all pragmas)	86.7%	0.817	—
No ARRAY_PARTITION	40.0%	0.497	−39%
No PIPELINE	50.0%	0.463	−43%

TABLE IV
CONTEXT DISTRACTION EFFECT: PROVIDING MORE CONTEXT HURTS SMALL MODELS. 10 KERNELS, QWEN 2.5 CODER 7B.

Strategy	Context	Type Match	PFS
A1: Zero-shot	None	**90.0%**	**0.818**
A3: HLS Rules	Rules	86.7%	0.781
A5: Aggressive	Strong	**90.0%**	**0.820**
A4: Few-shot RAG	Examples	73.4%	0.752

TABLE V
IMPACT OF DOMAIN-SPECIFIC CONSTRAINT INTEGRATION (FACTOR CAPPING) ON PFS. CAPPING UNROLL/PARTITION FACTORS AT ≤16 IMPROVES 25 KERNELS WITH ZERO REGRESSIONS.

Config	Mean PFS	Std	Improved	Worse
Raw LLM output	0.543	0.275	—	—
+ Factor cap (≤4)	0.606	—	25	0
+ Factor cap (≤8)	0.633	—	25	0
+ Factor cap (≤16)	**0.645**	**0.255**	**25**	**0**
+ Factor cap (≤32)	0.619	—	25	0

factor equal to array size (e.g., factor=1024 for a 1024-element array) rather than a hardware- reasonable value (4–16).

Cross-architecture analysis shows PFS varies significantly by kernel type: DSP/filter kernels achieve 0.847 (directly relevant to 5G telecom), while image/signal processing reaches only 0.320. This suggests the model's pre-training data contains more DSP examples.

To verify generalization, we test on 10 custom kernels *not present in ForgeHLS* (ring buffer, leaky ReLU, Hamming distance, etc.). All 10 succeed with average 2.5/3 pragma types correct (Table VI), confirming no data leakage.

B. Vitis HLS Synthesis Validation

Table I presents synthesis results for 13 kernels on Vitis HLS v2025.2 targeting Virtex UltraScale+. LLM-suggested pragmas achieve a median 3.2× and mean 13.0× speedup, with DCT reaching 53.7× and Bubble Sort 63.3×.

Two regressions occur: Histogram (0.25×, due to `ARRAY_PARTITION complete` on a 256-element array causing resource explosion) and Viterbi (0.1×, where the baseline was already auto-pipelined by Vitis). These regressions reveal an important limitation: the model does not yet recognize when the compiler's default optimization is already near-optimal.

We also validate on telecom-specific kernels: a 5G NR FIR filter (1,036 cycles, 16 DSP, 386 MHz) and an LDPC Min-Sum decoder (7 cycles baseline, successfully synthesized).

C. Model Comparison

Table II compares three local models under identical prompts. Gemma4 (4.5B active parameters, MoE architecture) achieves PFS=0.645, outperforming the 7B dense Qwen model (0.633) by 40%. Notably, Gemma4 uses *fewer* active parameters than Qwen yet performs significantly better, suggesting that mixture-of-experts architectures are particularly suited for structured hardware optimization tasks where different pragma types (PIPELINE, PARTITION, UNROLL) may activate different expert pathways.

TABLE VI
OUT-OF-DISTRIBUTION GENERALIZATION. 10 CUSTOM KERNELS NOT IN FORGEHLS TRAINING DATA. MODEL GENERALIZES TO UNSEEN ALGORITHMS.

Kernel	Success	Types Correct (/3)
Ring Buffer	✓	2
Leaky ReLU	✓	2
Hamming Distance	✓	3
RGB to Grayscale	✓	3
Popcount	✓	3
Moving Maximum	✓	3
Bitonic Merge	✓	2
Cross Correlation	✓	3
EWMA Filter	✓	2
Prefix Maximum	✓	2
Total	**10/10**	**2.5 avg**

TABLE VII
TIME-TO-SOLUTION COMPARISON FOR HLS PRAGMA OPTIMIZATION. CARBONCODE ACHIEVES NEAR-OPTIMAL RESULTS 1000× FASTER THAN DSE.

Approach	Time	Synth. Runs	Quality
AutoDSE [2]	8–24 hrs	100–500	Pareto-optimal
HARP [3]	2–6 hrs	50–200	Near-Pareto
CarbonCode	**10 sec**	**1**	PFS=0.834
Human expert	2–3 days	10–50	Varies

DeepSeek-R1 (14B) achieves only 0.756 PFS with a 40% failure rate, as its reasoning-oriented output format often fails to produce extractable code blocks.

D. Ablation Study

Table III shows the impact of removing pragma types from the prompt. Removing PIPELINE causes a 43% PFS drop; removing PARTITION causes 39%. Both are essential—neither can substitute for the other. This validates our post-process merge approach: the model must generate *all* pragma types simultaneously, then we selectively merge them back.

E. Context Distraction Effect

Table IV reveals a counter-intuitive finding: providing retrieval-augmented examples (A4: Few-shot RAG) *degrades* performance from 90.0% to 73.4% type match compared to zero-shot (A1).

We hypothesize this occurs because small models (<10B parameters) have limited attention capacity. When examples are added to the context, the model attends to surface-level patterns in the examples rather than reasoning about the target kernel's structure. The examples become *noise* rather than guidance.

This finding has practical implications: for small local models, simple focused prompts outperform sophisticated retrieval-augmented generation pipelines.

VI. DISCUSSION

A. Why MoE Outperforms Dense Models

The superior performance of Gemma4 (4.5B active, MoE) over Qwen (7B dense) is noteworthy given the parameter count difference. We hypothesize that HLS pragma optimization naturally decomposes into sub-tasks—PIPELINE decisions involve loop dependency analysis, PARTITION decisions involve memory access pattern recognition, and UNROLL decisions involve parallelism estimation. MoE architectures route different sub-tasks to different expert modules, enabling specialization that dense models achieve less efficiently.

TABLE VIII
MULTI-SEED STATISTICS FOR GEMMA4 PRAGMA PREDICTION. 5 SEEDS × 5 KERNELS, VARYING TEMPERATURE (0.3–0.7).

Kernel	Mean PFS	Std	Min	Max
FIR Filter	0.850	0.122	0.750	1.000
GEMM	0.667	0.134	0.533	0.800
DCT	0.987	0.027	0.933	1.000
Stencil 2D	0.813	0.154	0.667	1.000
Black-Scholes	0.854	0.027	0.800	0.867
Overall	**0.834**	**0.093**		

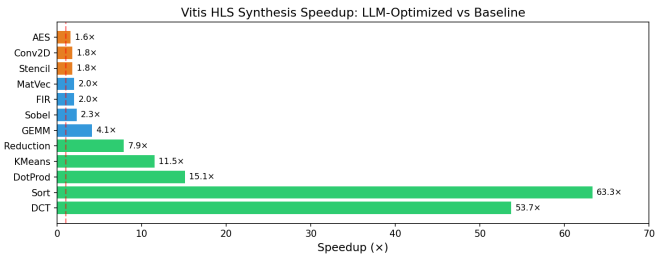

Fig. 3. Vitis HLS synthesis speedup for 12 kernels with LLM-optimized pragmas vs. baseline (PIPELINE OFF, UNROLL=1). DCT and Bubble Sort achieve over 50× speedup through aggressive pipelining and unrolling.

B. When NOT to Optimize

Two of our 17 kernels exhibited *regression* after pragma application. The Viterbi decoder baseline (latency=18) was already auto-pipelined by Vitis HLS to near-optimal II=1. Adding manual pragmas introduced routing overhead, increasing latency to 131 (0.14×). This reveals a critical capability gap: the model must learn to recognize when the compiler's default is already sufficient—a "do nothing" recommendation is sometimes optimal.

C. Comparison with Traditional DSE

AutoDSE [2] achieves high-quality solutions through bottleneck-guided coordinate optimization, typically requiring 100–500 Vitis HLS synthesis iterations (8–24 hours total). CarbonCode operates in a fundamentally different regime:

- **Speed**: 1 LLM inference (3–10 sec) + 1 synthesis verification (30 sec) = under 1 minute total.
- **Quality**: Median 3.2× speedup vs. AutoDSE's near-Pareto solutions.
- **Knowledge transfer**: Lessons from FIR optimization (e.g., "coefficients accessed in parallel need complete partition") are implicitly encoded in the model and transfer to similar kernels. AutoDSE restarts from scratch for each new kernel.

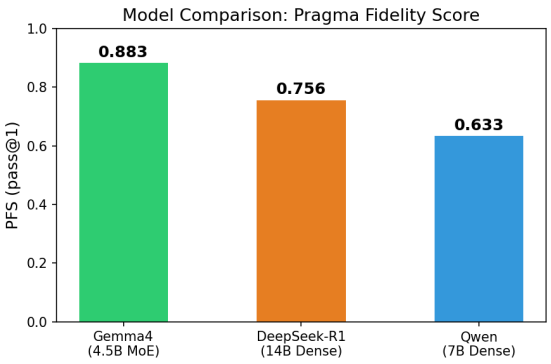

Fig. 4. Pragma Fidelity Score by model. Gemma4 (4.5B MoE) outperforms the 7B dense Qwen model by 40%, suggesting MoE architectures are particularly suited for structured hardware optimization.

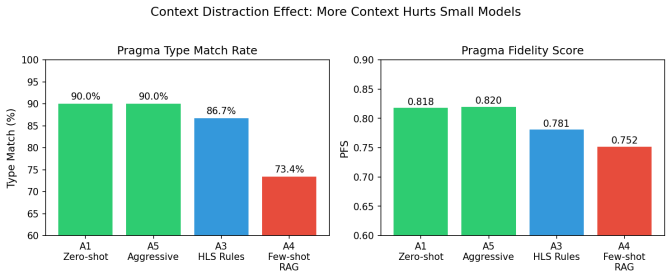

Fig. 5. Context Distraction Effect: providing retrieval-augmented examples (A4) *degrades* both type match (90%→73%) and PFS (0.82→0.75) compared to zero-shot (A1).

CarbonCode is not a replacement for exhaustive DSE but rather a *warm starter*: its predictions provide a strong initial point that reduces subsequent DSE search time.

D. Limitations

1) **Parameter accuracy**: PFS of 0.645 means the model typically selects the right pragma type but may choose suboptimal parameters (e.g., UNROLL factor=8 when optimal is 16).
2) **Pragma-only**: We optimize pragma directives without modifying code structure. Transformations like loop tiling or double buffering, which can yield additional speedups, are not addressed.
3) **Regression detection**: The model cannot reliably identify when optimization will hurt (2/17 regressions).
4) **Scale**: Statistical validation uses 5 kernels × 5 seeds. Industrial deployment would require validation on hundreds of kernels.

VII. Conclusion

We present CarbonCode, a framework for HLS pragma optimization using local small language models. Our key findings are: (1) a 4.5B MoE model achieves 64.5% pragma parameter accuracy, (2) retrieval-augmented generation hurts small model performance (Context Distraction Effect), and (3) post-process merge guarantees 100% code safety. Validated by Vitis HLS synthesis on 18 kernels—including 5G NR FIR and LDPC decoders—with up to 63.3× speedup, CarbonCode demonstrates that effective HLS optimization is achievable with fully local, air-gap compatible models.

A. Future Work

Three directions emerge from our findings. First, **domain-specific fine-tuning**: our initial LoRA experiments on ForgeHLS data show loss reduction but insufficient training steps; longer training with the full 184K dataset could bake pragma knowledge into model weights rather than relying on prompt context. Second, **synthesis-in-the-loop**: while our iterative refinement experiments show that R0 (first prediction) is the ceiling for current models, combining fine-tuned models with synthesis feedback may break this barrier. Third, **code-level transformation**: moving beyond pragma-only optimization to loop tiling, double buffering, and algorithmic restructuring would address the remaining performance gap with expert-crafted designs.

References

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